

MOBILE SIGINT PLATFORM

Technical Requirements Document & Product Requirements Document

*Vehicle-Mounted RF Intelligence Collection System
with HaLow Mesh Network Integration*

Version 1.0

January 2026

CONFIDENTIAL - FOR INTERNAL USE ONLY

Table of Contents

Table of Contents.....	2
1. Executive Summary	4
1.1 Key Capabilities	4
1.2 Target Users.....	4
2. System Architecture	5
2.1 Architecture Overview.....	5
2.2 Hardware Components	5
2.3 Network Architecture	5
2.4 IP Address Allocation.....	6
3. Functional Requirements	7
3.1 Scanner Control System (FR-100).....	7
3.1.1 Requirements.....	7
3.2 Direction Finding System (FR-200).....	7
3.2.1 Requirements.....	7
3.3 RF Monitoring System (FR-300)	7
3.3.1 Kismet Integration	7
3.3.2 ADS-B Monitoring	8
3.3.3 ISM Band Monitoring.....	8
3.4 Sensor Mesh Network (FR-400)	8
3.4.1 Requirements.....	8
3.5 HaLow Mesh Network (FR-500)	8
3.5.1 Requirements.....	9
3.6 Data Aggregation and Storage (FR-600)	9
3.7 User Interface (FR-700).....	9
4. Non-Functional Requirements.....	10
4.1 Performance Requirements (NFR-100)	10
4.2 Reliability Requirements (NFR-200)	10
4.3 Environmental Requirements (NFR-300).....	10
4.4 Security Requirements (NFR-400).....	10
4.5 Maintainability Requirements (NFR-500)	10
5. Software Stack.....	12
5.1 Operating Systems	12
5.2 Core Services by Host.....	12
5.2.1 Main Controller (Pi 5 #2)	12
5.2.2 Kraken Host (Pi 5 #1).....	12

5.2.3 ADS-B Station (Pi 4)	12
5.3 Development Stack	13
6. Hardware Specifications	14
6.1 Power System	14
6.2 Enclosure Requirements	14
6.3 Antenna Requirements	14
7. Data Model	15
7.1 MQTT Topic Hierarchy	15
7.2 Data Retention Policy	15
8. Integration Points	16
8.1 External APIs.....	16
8.1.1 Scanner Control API	16
8.1.2 KrakenSDR API	16
8.1.3 Aggregator API	16
8.2 Service Dependencies.....	16
9. Testing Requirements	17
9.1 Unit Testing	17
9.2 System Testing.....	17
9.3 Environmental Testing	17
10. Appendices	18
10.1 Glossary	18
10.2 References	18
10.3 Revision History.....	18

1. Executive Summary

This document defines the technical and product requirements for a comprehensive Mobile SIGINT (Signals Intelligence) Platform designed for vehicle-mounted deployment. The system integrates multiple RF sensing technologies, direction finding capabilities, and mesh networking to create a complete signals intelligence collection and analysis solution.

The platform addresses three primary operational scenarios: mobile RF surveillance with real-time direction finding, distributed sensor network coordination, and emergency communications via resilient mesh networking. The architecture leverages commodity hardware (Raspberry Pi, ESP32, RTL-SDR) combined with specialized equipment (KrakenSDR, Morse Micro HaLowLink) to achieve professional-grade capabilities at accessible price points.

1.1 Key Capabilities

- Multi-band RF monitoring: BLE, WiFi, ADS-B, TPMS, ISM band devices, trunked radio systems
- Real-time direction finding via KrakenSDR 5-channel coherent receiver
- Scanner integration with one-button DF handoff (Uniden BCD996P2)
- Distributed sensor mesh for wide-area coverage
- Dual-WAN connectivity: Starlink (primary) + HaLow mesh (backup)
- GridDown-ready operation with local services when internet unavailable
- Touch-optimized driving interface for mobile operation

1.2 Target Users

- Amateur radio operators conducting propagation studies and repeater mapping
- Security researchers performing RF spectrum analysis
- Emergency communications teams requiring resilient mesh networks
- Aviation enthusiasts tracking aircraft via ADS-B

2. System Architecture

2.1 Architecture Overview

The Mobile SIGINT Platform employs a modular, layered architecture with clear separation between sensing, processing, networking, and presentation concerns. All components communicate via standard protocols (TCP/IP, MQTT, HTTP REST) enabling future expansion and third-party integration.

2.2 Hardware Components

Device	Role	Specifications
Raspberry Pi 5 #1	KrakenSDR Host	8GB RAM, NVMe SSD, dedicated to DSP workload
Raspberry Pi 5 #2	Main Controller	8GB RAM, 256GB SSD, Kismet, aggregator, dashboard, WiFi AP
Raspberry Pi 4	ADS-B Station	4GB RAM, readsb, tar1090, optional AIS
HaLowLink (Vehicle)	Mesh Gateway	OpenWRT, 2.4GHz AP, 915MHz HaLow mesh
HaLowLink (Home)	Backup WAN	Internet gateway for mesh failover
Netgear GS308	Network Switch	8-port gigabit, 12V native input
Starlink	Primary WAN	Satellite internet, 50-200 Mbps
BCD996P2	Scanner	Trunked radio monitoring, serial CAT control
KrakenSDR	Direction Finding	5-channel coherent RTL-SDR array
Zero 2W (x6)	Remote Sensors	Distributed WiFi/BLE monitoring nodes
ESP32/nRF52	Edge Sensors	BLE beacons, WiFi probes, ISM monitoring

2.3 Network Architecture

The platform utilizes four distinct network segments, each serving a specific purpose:

Subnet	CIDR	Purpose
Vehicle Main LAN	192.168.1.0/24	Core infrastructure: Pis, switch, Starlink gateway
Guest WiFi	192.168.2.0/24	Isolated network for phones/tablets via HaLowLink AP
Sensor Mesh	10.0.0.0/24	Private network for Zero 2W and ESP32 sensors
HaLow Backbone	172.16.0.0/24	915MHz mesh linking vehicle to remote nodes and home base

2.4 IP Address Allocation

Device	IP Address	Network
Starlink Router	192.168.1.1	Vehicle LAN (GW)
Pi 5 Main Controller	192.168.1.50, 10.0.0.1	Vehicle LAN + Sensor
Pi 5 Kraken Host	192.168.1.51	Vehicle LAN
Pi 4 ADS-B Station	192.168.1.52	Vehicle LAN
HaLowLink (Vehicle)	192.168.1.53, 192.168.2.1, 172.16.0.1	All networks
HaLowLink (Home Base)	172.16.0.254	HaLow Backbone
Zero 2W Nodes	10.0.0.20-29	Sensor Mesh
ESP32 Nodes	10.0.0.30-39	Sensor Mesh

3. Functional Requirements

3.1 Scanner Control System (FR-100)

The scanner control system provides a touch-optimized interface for the Uniden BCD996P2 scanner, designed for safe operation while driving.

3.1.1 Requirements

- FR-101: System shall communicate with BCD996P2 via USB serial at 9600 baud
- FR-102: System shall display current frequency in large, high-contrast format (minimum 12vw font)
- FR-103: System shall display system name and channel name for trunked systems
- FR-104: System shall provide touch targets minimum 48x48 pixels for all controls
- FR-105: System shall support Hold, Scan, and Lockout functions via touch interface
- FR-106: System shall maintain channel history (minimum 50 entries) with timestamps
- FR-107: System shall provide one-button handoff to KrakenSDR for direction finding
- FR-108: System shall auto-start on boot and run in kiosk mode

3.2 Direction Finding System (FR-200)

The direction finding system leverages the KrakenSDR 5-channel coherent receiver to determine bearing to RF emitters.

3.2.1 Requirements

- FR-201: System shall accept frequency input from scanner control system via REST API
- FR-202: System shall tune KrakenSDR to specified frequency within 500ms
- FR-203: System shall display bearing with minimum 5-degree accuracy
- FR-204: System shall display confidence level for bearing estimate
- FR-205: System shall integrate GPS for georeferenced bearing lines
- FR-206: System shall log all DF measurements with timestamp, position, frequency, and bearing
- FR-207: System shall support bearing overlay on map display

3.3 RF Monitoring System (FR-300)

The RF monitoring system provides continuous surveillance of multiple frequency bands and protocols.

3.3.1 Kismet Integration

- FR-301: System shall run Kismet server on Main Controller
- FR-302: System shall support WiFi monitoring in 2.4GHz and 5GHz bands

- FR-303: System shall support BLE device detection and tracking
- FR-304: System shall support remote capture sources from Zero 2W nodes
- FR-305: System shall maintain device database with first-seen, last-seen timestamps
- FR-306: System shall export data in Kismet, Wigle, and CSV formats

3.3.2 ADS-B Monitoring

- FR-311: System shall receive and decode ADS-B signals at 1090 MHz
- FR-312: System shall display aircraft on map with track history
- FR-313: System shall show aircraft callsign, altitude, speed, and heading
- FR-314: System shall calculate distance and bearing to aircraft from current position
- FR-315: System shall support MLAT for non-ADS-B aircraft (when networked)

3.3.3 ISM Band Monitoring

- FR-321: System shall decode ISM band devices via rtl_433
- FR-322: System shall identify and log TPMS sensors with tire position
- FR-323: System shall decode weather station transmissions
- FR-324: System shall detect and log wireless doorbells, sensors, and remotes
- FR-325: System shall publish decoded data to MQTT broker

3.4 Sensor Mesh Network (FR-400)

The sensor mesh extends monitoring capability beyond the vehicle through distributed sensor nodes.

3.4.1 Requirements

- FR-401: Main Controller shall operate WiFi access point for sensor mesh (SSID: SensorMesh)
- FR-402: System shall support minimum 20 concurrent sensor connections
- FR-403: System shall provide DHCP and DNS services to sensor network
- FR-404: System shall run MQTT broker for sensor data ingestion
- FR-405: Sensor nodes shall report via MQTT with QoS 1 minimum
- FR-406: System shall support multiple transport paths (WiFi, HaLow) for redundancy
- FR-407: Sensor data shall include device ID, timestamp, GPS coordinates (if available), and payload

3.5 HaLow Mesh Network (FR-500)

The HaLow mesh provides long-range, low-bandwidth connectivity for extended operations and WAN failover.

3.5.1 Requirements

- FR-501: Vehicle HaLowLink shall operate as mesh node on 915 MHz
- FR-502: Vehicle HaLowLink shall provide 2.4 GHz guest AP (isolated from sensor mesh)
- FR-503: System shall maintain mesh routing table for multi-hop paths
- FR-504: System shall support automatic WAN failover from Starlink to HaLow path
- FR-505: Failover shall occur within 30 seconds of primary WAN failure
- FR-506: System shall fail back to Starlink when available (with hysteresis)
- FR-507: HaLow network shall support minimum 1 Mbps throughput at 500m range
- FR-508: Mesh shall operate with minimum 3 nodes for redundancy

3.6 Data Aggregation and Storage (FR-600)

- FR-601: System shall store time-series data in InfluxDB
- FR-602: System shall retain minimum 30 days of operational data locally
- FR-603: System shall support data export to USB storage
- FR-604: System shall correlate data across sources (scanner, Kismet, ADS-B, sensors)
- FR-605: System shall generate unified event timeline
- FR-606: System shall support real-time data streaming to external systems

3.7 User Interface (FR-700)

- FR-701: System shall provide unified web dashboard accessible at main-controller.local
- FR-702: Dashboard shall be optimized for 7-inch touchscreen (1024x600)
- FR-703: Dashboard shall support dark mode for night operation
- FR-704: Dashboard shall display real-time status of all subsystems
- FR-705: Dashboard shall provide quick-access links to Kismet, tar1090, Kraken UIs
- FR-706: Dashboard shall display current GPS position and heading
- FR-707: Dashboard shall show WAN status and current route (Starlink vs HaLow)

4. Non-Functional Requirements

4.1 Performance Requirements (NFR-100)

- NFR-101: Scanner UI shall update frequency display within 300ms of change
- NFR-102: Kraken tuning shall complete within 500ms of command
- NFR-103: DF bearing shall update at minimum 2 Hz
- NFR-104: Dashboard shall load within 3 seconds on local network
- NFR-105: MQTT message latency shall be under 100ms on sensor mesh
- NFR-106: System shall process minimum 1000 sensor messages per second

4.2 Reliability Requirements (NFR-200)

- NFR-201: System shall operate continuously for minimum 8 hours without restart
- NFR-202: System shall recover automatically from transient USB disconnections
- NFR-203: Services shall restart automatically on failure (systemd watchdog)
- NFR-204: Data shall be persisted to survive unexpected power loss
- NFR-205: WAN failover shall be transparent to running applications

4.3 Environmental Requirements (NFR-300)

- NFR-301: System shall operate in ambient temperatures from 0°C to 45°C
- NFR-302: System shall tolerate vehicle vibration per SAE J1455
- NFR-303: Pelican case shall provide IP67 protection when sealed
- NFR-304: System shall operate from 12V vehicle power (10.5V - 14.4V range)
- NFR-305: Total system power consumption shall not exceed 100W

4.4 Security Requirements (NFR-400)

- NFR-401: Sensor mesh WiFi shall use WPA2-PSK minimum
- NFR-402: HaLow mesh shall use WPA3-SAE encryption
- NFR-403: Guest WiFi shall be isolated from all internal networks
- NFR-404: Web interfaces shall be accessible only from local networks
- NFR-405: MQTT broker shall require authentication
- NFR-406: SSH access shall use key-based authentication only

4.5 Maintainability Requirements (NFR-500)

- NFR-501: All software shall be deployable via Ansible playbooks
- NFR-502: Configuration shall be version-controlled in Git
- NFR-503: System shall support OTA updates when internet available

- NFR-504: Logs shall be accessible via journalctl and web interface
- NFR-505: Hardware shall be accessible for maintenance without disassembly

5. Software Stack

5.1 Operating Systems

Device	OS	Notes
Pi 5 (Main)	Raspberry Pi OS Lite (64-bit)	Bookworm-based, headless
Pi 5 (Kraken)	Raspberry Pi OS Lite (64-bit)	KrakenSDR image preferred
Pi 4 (ADS-B)	DietPi or Pi OS Lite	Lightweight for single purpose
HaLowLink	OpenWRT 23.05+	Morse Micro driver support
Zero 2W	Raspberry Pi OS Lite (32-bit)	Minimal install

5.2 Core Services by Host

5.2.1 Main Controller (Pi 5 #2)

- scanner-ui: Custom Flask application for BCD996P2 control
- kismet: WiFi/BLE/RF monitoring and device tracking
- gpsd: GPS daemon providing time and position
- hostapd: WiFi access point for sensor mesh
- dnsmasq: DHCP and DNS for sensor network
- mosquitto: MQTT broker for sensor data
- influxdb: Time-series database
- grafana: Dashboard and visualization
- nginx: Reverse proxy for all web services
- aggregator: Custom service correlating all data sources

5.2.2 Kraken Host (Pi 5 #1)

- krakensdr_doa: Direction of arrival DSP processing
- krakensdr_pr: Passive radar (optional)
- nginx: Proxy for remote access

5.2.3 ADS-B Station (Pi 4)

- readsb: ADS-B decoder (replaces dump1090)
- tar1090: Web-based aircraft map
- graphs1090: Statistics and graphs
- mlat-client: Multilateration client

5.3 Development Stack

- Python 3.11+: Primary application language
- Flask: Web framework for custom services
- pyserial: Scanner serial communication
- paho-mqtt: MQTT client library
- influxdb-client: Database access
- pynmea2: GPS NMEA parsing
- requests: HTTP client for inter-service communication

6. Hardware Specifications

6.1 Power System

All components operate from vehicle 12V power via appropriate DC-DC converters.

Component	Voltage	Max Current	Converter
Pi 5 (x2)	5V	5A each	12V→5V 5A (x2)
Pi 4	5V	3A	12V→5V 3A
Netgear GS308	12V	0.5A	Direct 12V
HaLowLink	5V	2A	12V→5V 3A
7" Display	5V	1A	Shared with Pi 5
KrakenSDR	5V	2.5A	USB from Pi 5
Cooling Fans	5V/12V	0.5A	Various

Total estimated power consumption: 75-85W typical, 100W peak

6.2 Enclosure Requirements

- Pelican case with custom foam insert for component mounting
- Panel-mount connectors: 2x RJ45 (Ethernet pass-through), 2x USB-C (power), N-type (antennas)
- Ventilation: Active cooling with filtered intake, exhaust fan
- Internal temperature monitoring with thermal shutdown protection
- Accessible without full disassembly for SD card and USB access

6.3 Antenna Requirements

Purpose	Frequency	Type	Mounting
KrakenSDR DF	Various	5x matched dipole array	Roof magnetic mount
ADS-B	1090 MHz	Vertical collinear	Roof magnetic mount
HaLow	915 MHz	Omni or directional	Roof or case mount
WiFi Sensor	2.4/5 GHz	Dual-band omni	Internal or case mount
GPS	1575 MHz	Active patch	Roof magnetic mount

7. Data Model

7.1 MQTT Topic Hierarchy

All sensor data flows through MQTT using a hierarchical topic structure:

- sensors/wifi/probes - WiFi probe requests from ESP32 nodes
- sensors/wifi/devices - Detected WiFi devices from Kismet
- sensors/ble/advertisements - BLE advertisements from nRF52 nodes
- sensors/ble/devices - Processed BLE device records
- sensors/ism/rtl433 - Decoded ISM band devices
- sensors/ism/tpms - Tire pressure monitor data
- scanner/status - Current scanner state (frequency, system, channel)
- scanner/history - Channel activity log
- kraken/bearing - Current DF bearing and confidence
- kraken/target - Active DF target information
- adsb/aircraft - Aircraft position updates
- system/gps - Vehicle position and heading
- system/status - Health and status of all subsystems
- system/alerts - System alerts and notifications

7.2 Data Retention Policy

Data Type	Retention	Storage
Raw sensor messages	7 days	InfluxDB
Aggregated device records	30 days	InfluxDB
DF measurements	90 days	InfluxDB + SQLite
Scanner history	30 days	SQLite
Kismet database	Per session	SQLite (kismetdb)
System logs	14 days	journald

8. Integration Points

8.1 External APIs

8.1.1 Scanner Control API

REST API exposed by scanner-ui service on port 8080:

- GET /api/status - Current scanner state
- POST /api/hold - Toggle hold on current frequency
- POST /api/scan - Resume scanning
- POST /api/lockout - Lock out current channel
- GET /api/history - Recent channel history
- POST /api/kraken/tune - Hand off frequency to KrakenSDR

8.1.2 KrakenSDR API

REST/WebSocket API provided by KrakenSDR software on port 8080:

- POST /settings - Configure frequency, gain, DOA parameters
- GET /doa_data - Current bearing and confidence
- WebSocket /ws - Real-time DOA stream

8.1.3 Aggregator API

Central data ingestion endpoint on port 5000:

- POST /ingest - Accept sensor data from any source
- GET /devices - Query device database
- GET /timeline - Event timeline with filters
- GET /export - Data export in various formats

8.2 Service Dependencies

Service startup order and dependencies are managed by systemd:

- gpsd must start before scanner-ui and aggregator
- mosquitto must start before aggregator and sensor services
- influxdb must start before aggregator and grafana
- nginx starts last after all backend services

9. Testing Requirements

9.1 Unit Testing

- All Python modules shall have minimum 80% code coverage
- Serial communication shall be testable with mock scanner
- MQTT handling shall be tested with in-memory broker
- API endpoints shall have integration tests

9.2 System Testing

- End-to-end scanner control workflow
- Scanner to Kraken handoff timing verification
- Sensor data flow from edge to dashboard
- WAN failover and recovery scenarios
- Power cycle recovery verification

9.3 Environmental Testing

- Thermal testing at 0°C and 45°C ambient
- Vibration testing during vehicle operation
- Extended operation testing (8+ hours continuous)
- Power variation testing (10.5V - 14.4V)

10. Appendices

10.1 Glossary

- ADS-B: Automatic Dependent Surveillance-Broadcast - aircraft transponder system
- BLE: Bluetooth Low Energy
- CAT: Computer Aided Transceiver control
- DF: Direction Finding
- DOA: Direction of Arrival
- HaLow: IEEE 802.11ah - sub-GHz WiFi standard
- ISM: Industrial, Scientific, Medical - unlicensed radio bands
- MQTT: Message Queuing Telemetry Transport
- SDR: Software Defined Radio
- SIGINT: Signals Intelligence
- TPMS: Tire Pressure Monitoring System

10.2 References

- KrakenSDR Documentation: https://github.com/krakenrf/krakensdr_docs
- Kismet Documentation: <https://www.kismetwireless.net/docs/>
- Uniden BCD996P2 Serial Protocol: Owner's Manual Appendix
- Morse Micro HaLowLink: <https://www.morsemicro.com/>
- OpenWRT Documentation: <https://openwrt.org/docs/>
- readsb ADS-B Decoder: <https://github.com/wiedehopf/readsb>

10.3 Revision History

Version	Date	Author	Changes
1.0	2026-01-18	Initial	Initial release